

PoR-Chain: Cryptographically Anchored Cumulative Hash Chains for Provable Multimedia Authenticity

Sungho Kim
KANT Labs
sh@kantlabs.com

May 18, 2026

Abstract

The proliferation of generative AI has eroded public trust in multimedia content, with deepfakes increasingly used in financial fraud, political disinformation, and fabricated evidence. Existing provenance frameworks such as the Coalition for Content Provenance and Authenticity (C2PA) [2] rely on metadata signing schemes that fail to verify capture authenticity once a trusted device is compromised, while sensor fingerprinting techniques such as Photo Response Non-Uniformity (PRNU) [3] are vulnerable to trained adversaries. We present **PoR-Chain**, a cryptographic framework that binds multimedia content to a specific moment in time and a specific physical capture device through a physical feedback path that no software-only adversary can reproduce. The foundation—*Proof-of-Reality (PoR)*—embeds time-varying data sourced from a distributed ledger into multimedia content via an optical, acoustic, or electrical signal path during capture; the resulting content cannot be synthesized in advance because the embedded signal would not match the ledger state at any point prior to capture. PoR-Chain anchors a single initial blockhash and generates a SHA-256 chain that is encoded sequentially into successive content groups, providing both single-query verification and tamper-evident sequence integrity for continuous video. Bidirectional time bounding, achieved by recording a termination anchor on the ledger, constrains the time window in which content could have been generated and breaks the economic feasibility of post-hoc synthesis attacks. We describe a working implementation using Solana and a transparent OLED panel, and analyze adversarial scenarios including deepfake injection, frame splicing, lipsync forgery, and AI voice cloning. The PoR framework is described in full in a companion paper [1]; this paper focuses on PoR-Chain and its integration with PoR for continuous video.

Keywords: proof-of-reality, content provenance, deepfake detection, physical unclonable functions, blockchain anchoring, hash chains, multimedia authentication

1 Introduction

Generative AI has eroded trust in multimedia content faster than forensic detection can keep pace. The companion paper [1] surveys existing provenance approaches—device-signing schemes, sensor fingerprinting, watermarking, and blockchain timestamping—and shows that none verifies the *act of capture* itself. That paper introduces the Proof-of-Reality (PoR) framework, which closes this gap for still images by binding content to a live ledger signal through a physical feedback path.

Hash chains. Cryptographic hash chains as one-way commitment structures date to Lamport [4]. PoR-Chain uses them as a sequence integrity primitive bound to a ledger state, providing tamper-evident ordering across content groups with a single ledger query—distinguishing this application from traditional uses in authentication tokens.

This paper extends PoR to continuous video through **PoR-Chain**. The extension rests on two complementary insights:

1. **Physical feedback paths cannot be simulated retroactively.** If content must embed a signal broadcast through a physical channel and re-captured by the same device, no purely software adversary can synthesize valid content—because the signal depends on data that was not knowable before capture.
2. **Hash chains anchored to a single ledger state provide efficient sequence integrity.** A single anchor blockhash seeds a SHA-256 chain encoded into successive content groups. Verification requires a single ledger query and local recomputation; any deletion, reordering, or substitution of content groups breaks the chain.

2 PoR-Chain Framework

2.1 PoR Framework

The PoR framework is described in full in the companion paper [1]. In brief: during capture, time-varying data $D(t)$ —specifically, the Solana blockhash for the active slot, unpredictable before time t —is broadcast through a physical signal path (optical, acoustic, or electrical) and re-captured by the same device recording the content. The captured content thus embeds $D(t)$ at the moment of capture. Verification confirms that the embedded signal matches the ledger and that the on-chain receipt was submitted within a configurable freshness window W of the embedded slot. W is the key security parameter: pre-synthesis is defeated by the unpredictability of $D(t)$, and real-time synthesis is defeated as long as the adversary cannot synthesize and submit a valid receipt within W .

Two limitations of the base framework motivate PoR-Chain:

1. **Transaction cost.** Each content group must be individually anchored on the blockchain, requiring as many on-chain transactions as there are captured groups. This scales linearly with content duration and is prohibitively expensive for long recordings.
2. **Sequence integrity.** Detecting deletion or reordering of content groups requires comparing decoded signals against an independently retrieved sequence of $D(t)$ values, which is brittle under network jitter or partial decoding failures.

2.2 PoR-Chain Extension

PoR-Chain addresses both transaction cost and sequence integrity by deriving a cryptographic chain from the blockhash H_0 already fetched by PoR’s ledger challenge.

2.2.1 Chain Generation

Using H_0 as the seed, the device maintains a single running SHA-256 hasher updated with each successive content group:

$$H_n = \text{SHA256}(H_0 \parallel F_1 \parallel F_2 \parallel \dots \parallel F_n) \quad (1)$$

where F_n denotes the n -th content group. For multimedia content, $F_n = V_n \| A_n$ is the concatenation of the synchronized video frame group V_n and audio chunk A_n ; for single-channel content, F_n is the frame group or audio chunk alone. The hash is computed incrementally without re-hashing prior state, so each H_n requires only one additional block operation. Any modification to F_k for $k \leq n$ changes H_n and all subsequent states.

2.2.2 Encoding into Content Groups

Because H_n is computed from the bytes of F_n , it cannot be embedded in F_n itself. Instead, H_n is encoded into a transmittable signal—an optical pattern displayed on a transparent OLED panel, a near-ultrasonic audio waveform, a modulation of the camera’s analog gain, or a similar physical signal—and emitted during the capture of a future content group F_{n+i} , where $i \geq 1$ is a positive integer determined by the capture pipeline (e.g., the number of frames required between OLED-on and OLED-off phases to allow ambient-light subtraction). The constraint $i \leq W$ ensures the embedding remains within the freshness window. The signal is re-captured by the same device, embedding H_n within F_{n+i} . In practice, only the leading 128 bits of the 256-bit H_n are encoded, constrained by the physical signal capacity (e.g., the pixel count of the OLED display). Verification accepts a decoded value if its Hamming distance from the expected H_n is within a fixed threshold `FUZZY_BITS`; the collision probability of this fuzzy check is analysed in Section 4.8.

2.2.3 Bidirectional Time Bounding

After capture completes, the device posts the file hash of the captured content as a *termination anchor* on the ledger. Let T_R denote the time of the confirmed ledger slot S_R for this transaction.

A verifier checks that:

$$T_R - T_0 \leq T_{\text{playback}} + W \quad (2)$$

where T_{playback} is the duration of the captured content and W is the freshness window defined in Section 2.1. If $T_R - T_0$ greatly exceeds T_{playback} , the content was either generated in advance (impossible because H_0 would not have been knowable) or generated post-hoc (contradicting the termination time). The construction thus bounds the synthesis window bidirectionally.

Crucially, W is a fixed system constant determined by network round-trip time and ledger slot interval, whereas T_{playback} grows with content length. An adversary must synthesize T_{playback} seconds of content within $T_{\text{playback}} + W$ seconds, leaving only W seconds of slack regardless of content duration. When $T_{\text{playback}} \gg W$, the adversary is therefore constrained to synthesize at real-time speed or faster—a requirement that current diffusion-based video models cannot meet. Longer recordings thus provide strictly stronger security guarantees under this bound.

2.2.4 Multi-Channel Encoding (Video + Audio)

For multimedia content, each H_n is derived from the joint content group $F_n = V_n \| A_n$, so the chain cryptographically binds the video and audio channels together. The same H_n is then encoded into both channels simultaneously: the optical signal embeds H_n in the video frame group V_{n+i} , while the acoustic signal embeds H_n in the audio chunk A_{n+i} .

When the acoustic channel is implemented, this construction will provide cryptographic detection of:

- **Lipsync forgery:** an adversary substituting a synthetic audio track over genuine video would produce content in which the video and audio chains do not align, or in which the audio chain fails to match the ledger anchor.

- **Voice cloning attacks:** synthesized audio paired with genuine video would fail the audio chain check.
- **Deepfake video attacks:** synthesized video paired with genuine audio would fail the video chain check.

If verification of one channel fails (e.g., due to compression-induced decoding errors), the other channel can serve as a fallback, providing defense in depth. Acoustic encoding implementing this design is planned for future work.

2.2.5 Verification Protocol

Algorithm 1 PoR-Chain verification

- 1: Query ledger: $H_0 \leftarrow \text{blockhash}(S_0)$.
 - 2: Locally compute $H_1, H_2, \dots, H_N$ from H_0 via Equation 1.
 - 3: Decode embedded values H'_n from F_{n+i} for all n , where i is the known pipeline offset.
 - 4: Verify $H_n \approx H'_n$ for all n (Hamming distance $\leq \text{FUZZY_BITS}$).
 - 5: Compute $\Delta T \leftarrow \text{time}(S_R) - \text{time}(S_0)$.
 - 6: Verify $\Delta T \leq T_{\text{playback}} + W$.
 - 7: Accept if all checks pass; reject otherwise.
-

3 Implementation

We describe the current reference implementation, which realizes PoR-Chain (Equation 1). Acoustic encoding, dual-channel video-audio binding, and the standalone verifier CLI are not yet implemented; they are described as future work in Section 5.3.

3.1 Hardware

The reference device combines a single-board computer, a camera module, and a transparent OLED panel mounted in front of the lens. The transparent OLED is the key hardware choice: it allows the camera to capture both the real-world scene and the hash-chain signal in the same frame, without occluding the scene. A Waveshare 1.51-inch transparent OLED (128×64 pixels) serves as the display in the reference build.

The camera alternates lens focus between the OLED panel (close range) and the scene (normal distance). Frames captured during focus transitions are excluded from chain decoding, as the OLED grid is not reliably readable during refocus. This focus-switching cadence is the primary contributor to the pipeline offset i introduced in Section 2.2.

3.2 Software

Key components:

- **Anchor acquisition:** A WebSocket client subscribes to a Solana [6] broadcaster service and decodes the current blockhash H_0 at recording start.
- **Chain generation:** An incremental SHA-256 hasher is seeded with H_0 and updated continuously with the H.264 bitstream. At each OLED tick (33 ms), the current chain state is snapshotted without disturbing the ongoing computation.

- **Optical encoding:** The first 128 bits of the current hash state are rendered as a 16×8 grid of cells on the transparent OLED panel (one cell per bit). The display alternates between ON (≈ 165 ms) and OFF (33 ms) phases to enable ambient-light subtraction during decoding.
- **Termination anchor:** At recording end, the file hash of the captured content is posted to Solana via a signed RPC transaction. The confirmed slot S_R is saved alongside the video as a sidecar file.

3.3 Verification Pipeline and Empirical Decoding Performance

A verifier performs two independent checks on the recorded content:

1. **Exact hash check.** Recomputes $\text{SHA-256}(H_0 \parallel \text{entire video stream})$ in 4096-byte strides and compares it against the `.sha256` sidecar saved at shutdown.
2. **Visual OLED check.** Detects OLED ON/OFF blink cycles in the video frames, decodes the 128-bit hash state from each ON cycle using ambient subtraction, and fuzzy-matches the decoded values against the expected chain states. The check passes if at least half of the decoded cycles match (minimum 3 cycles required).

Empirical decoding accuracy. Figure 1 shows representative debug frames from a reference recording. Across 10 sampled blink cycles the per-cycle bit error count ranged from 1 to 6 out of 128 bits (mean 4.1 bits, 3.2 % bit error rate (BER)); all 10 cycles passed the visual check. The observed errors arise from systematic geometric misalignment between the OLED panel and the camera image plane, which shifts the decoded grid by a sub-cell amount. This is a calibration artifact rather than random optical noise; improving the physical mount reduces errors further. Regardless, the 8-bit fuzzy threshold absorbs the full observed error range, and the temporal averaging across multiple frames per blink cycle means no channel coding (e.g., Reed-Solomon) is needed in controlled capture environments. The security implications of this tolerance are quantified in Section 4.8.

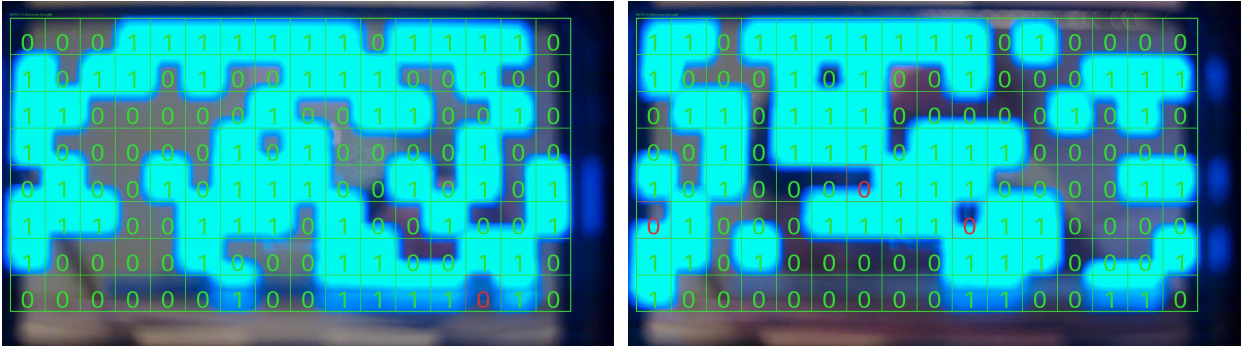


Figure 1: Two blink cycles from the same recording. The 16×8 grid is overlaid on the camera’s view of the transparent OLED panel; green cells are correctly decoded bits, red cells are errors. The outer border is green when the decoded 128-bit value fuzzy-matches (≤ 8 bit errors) the expected hash chain state, red otherwise. **Left:** best observed cycle (1/128 bit errors, 0.8 % BER). **Right:** typical cycle (3/128 bit errors, 2.3 % BER). Both pass. Errors cluster at cell-boundary columns, consistent with a fixed sub-cell OLED-to-camera geometric offset rather than random optical noise.

4 Security Analysis

Adversary capabilities. We assume an adversary who can generate photorealistic synthetic video and audio, splice or reorder frames, read public ledger state at any time, and has full knowledge of the protocol. Computational resources are unbounded for offline synthesis but bounded for real-time synthesis (discussed in Section 2.2.3).

Adversary goals. The adversary wishes to produce content that falsely claims to depict a real event, or that contains injected synthetic segments (deepfaked face, cloned voice) within otherwise genuine content, with the edit undetectable.

Trust assumptions. We assume ledger integrity (past blockhashes cannot be rewritten; future ones cannot be predicted), sound cryptographic primitives (SHA-256 [5]), and that the capture device is not under adversarial physical control during capture. We additionally assume the anchor broadcaster supplies authentic blockhashes; a device should confirm H_0 against a direct Solana RPC query before beginning capture. PoR-Chain provides defense in depth that supplements, rather than replaces, hardware roots of trust.

We analyze representative attack scenarios against PoR-Chain.

4.1 Pure Deepfake Injection

Attack: the adversary synthesizes content from scratch using a generative model and submits it as if captured.

Defense: synthesized content does not contain any embedded signal. Verification fails at step 3 of Algorithm 1 (decoded signal absent or invalid).

4.2 Frame Splicing

Attack: the adversary takes genuine PoR-Chain content and removes, reorders, or substitutes a subset of frames.

Defense: the cumulative hash chain breaks at the splice boundary. Verification fails at step 4.

4.3 Pre-Synthesis with Replay

Attack: the adversary observes the ledger state at S_0 , synthesizes content offline, then submits it claiming capture at $T_0 = \text{time}(S_0)$.

Defense: unless the adversary can synthesize and submit a termination anchor within $T_{\text{playback}} + W$ of T_0 , bidirectional time bounding fails. Current diffusion-based video models require significantly more than real-time compute per output-second on available hardware, making this attack infeasible for content longer than a few seconds; however, this margin is expected to erode as generative models improve.

4.4 Lipsync Forgery

Attack: the adversary takes genuine PoR-Chain video and overlays a synthesized voice track.

Defense (designed): when dual-channel encoding is implemented, the audio channel chain would mismatch the video channel chain (or fail verification against H_0 entirely). Acoustic encoding implementing this defense is planned for future work.

4.5 Voice Cloning Paired with Genuine Video

Attack: adversary clones a target’s voice, attaches it to a genuine PoR-Chain video, and submits.

Defense (designed): when dual-channel encoding is implemented, the audio chain check would fail identically to the lipsync forgery case.

4.6 Optical Signal Replay Attack

Attack: adversary records the OLED-emitted signal during a genuine capture, then plays it back during a separate synthetic capture.

Defense: because H_0 is unique to each capture session and the chain is deterministic from H_0 , replay of an old signal embeds an old chain. Bidirectional time bounding (Section 2.2.3) detects the mismatch between the claimed capture time and the actual ledger anchor time.

4.7 Screen Relay Attack

Attack: the adversary generates a synthetic scene using AI, plays it on a screen, and places the screen in front of the PoR-Chain capture device. The camera records the screen, which appears to be a real-world scene; the OLED hash signal is overlaid and re-captured as normal. The resulting video passes all chain and time bounding checks, because the physical feedback path requirement is genuinely satisfied.

Defense: PoR-Chain does not currently detect this attack. The chain verifies that the hash signal was physically re-captured, but cannot distinguish a real three-dimensional scene from a flat screen displaying synthetic content. Candidate mitigations include depth sensing (ToF, structured light, or stereo vision) to confirm that the scene has physical three-dimensional extent, and Moiré pattern detection to identify the pixel-grid interference artifacts that appear when a camera photographs a digital display. Neither mitigation is implemented in the reference build; both are directions for future work.

4.8 Fuzzy-Matching Collision Probability

The visual OLED check accepts a decoded 128-bit hash state if it falls within Hamming distance 8 of any entry in the hash chain (`FUZZY_BITS` = 8). This tolerance accommodates the systematic geometric decoding errors observed in practice (1–6 bit errors per cycle; see Section 3). However, it raises the question of whether the threshold materially weakens forgery resistance.

The probability that a uniformly random 128-bit string falls within Hamming distance 8 of a fixed target is:

$$P_{\text{fuzzy}} = \frac{\sum_{k=0}^8 \binom{128}{k}}{2^{128}} = \frac{1,529,927,642,833}{2^{128}} \approx 4.5 \times 10^{-27} \approx 2^{-87.5}.$$

This corresponds to an effective security level of 87.5 bits for the visual check taken in isolation—a loss of roughly 40.5 bits relative to an exact match, yet still far beyond any feasible brute-force budget.

Multi-cycle compounding. The visual check does not verify a single blink cycle: it verifies *every* OLED-on cycle throughout the recording. Each snapshot H_n is a deterministic function of H_0 and the preceding video bytes, so fooling all cycles simultaneously requires finding a video whose

streaming SHA-256 snapshots all fall within 8 bits of the legitimate chain—a problem equivalent to second-preimage resistance of SHA-256. As intuition for the per-cycle difficulty under random guessing, the probability that a uniformly random 128-bit string passes a single fuzzy check is $P_{\text{fuzzy}} \approx 4.5 \times 10^{-27} \approx 2^{-87.5}$; because the chain states are not independently guessable, the joint probability across N cycles is not simply P_{fuzzy}^N but is instead bounded by the SHA-256 second-preimage hardness above. The only computationally plausible approach is *real-time synthesis*: generating content while physically displaying the correct hash sequence on a hardware OLED during capture, which is precisely the attack class the physical feedback path is designed to make infeasible.

Conjunctive check. The visual check is conjunctive with Check 1 (exact SHA-256 of the full compressed video stream), which provides 256-bit preimage resistance independently. An adversary must defeat both checks simultaneously, so the combined security is bounded below by the stronger of the two.

4.9 Comparison with Prior Work

Table 1: Security property comparison

Property	C2PA	PRNU	Watermarking	Timestamp	PoR-Chain
Resists post-hoc synthesis	~	~	×	×	✓
Resists frame splicing	~	~	~	×	✓
Resists lipsync forgery	~	×	~	×	~
Single-query verification	✓	n/a	✓	✓	✓
Bidirectional time binding	×	×	×	×	✓
No hardware root of trust	×	✓	~	✓	✓*

*PoR-Chain does not require a trusted signing key but does require a capture device that supports physical signal emission and re-capture. ~ denotes partial or conditional resistance. For C2PA, ~ indicates that post-hoc attacks are detected (the content hash in the manifest breaks), but capture-time attacks—such as recording a screen displaying synthetic content—are not.

5 Discussion and Future Work

5.1 Standardization

PoR-Chain is complementary to existing standards. The framework can be expressed as a C2PA assertion type, with the initial anchor H_0 and termination slot recorded in the manifest. Adoption by C2PA, JPEG Trust, or a similar consortium would enable interoperable verification.

5.2 Privacy Considerations

Content privacy is preserved by design: only the file hash of the captured content and the anchor slot number S_0 are recorded on-chain under the user’s public key. The file hash is a one-way digest that reveals nothing about the video content. H_0 is already a publicly known Solana blockhash and discloses nothing beyond what the ledger already publishes.

Two residual metadata exposures remain. First, recording S_0 on-chain reveals the approximate capture time to within one Solana slot (≈ 400 ms); this is an inherent consequence of the temporal-anchoring design and is acceptable in most provenance use cases where capture time is intended to

be verifiable. Second, the submitting public key pseudonymously links multiple recordings to the same author; users who require stronger unlinkability may submit termination anchors through a privacy-preserving relay or use a fresh keypair per session.

5.3 Limitations

Hardware requirement. The framework requires a capture device with both signal emission and signal re-capture capabilities. Standalone software implementations are not possible.

Network dependency at beginning and end. Network connectivity is required only at the beginning of a recording—to fetch the initial anchor H_0 —and at the end, to post the termination anchor. No connection is needed during capture.

Partial optical encoding. The current OLED codec encodes only the first 128 bits of the 256-bit hash state, constrained by the 128×64 -pixel display. A higher-resolution display or an alternating two-frame scheme would expose the full 256 bits.

Controlled-environment assumption. Decoding accuracy has been characterized in a laboratory setting (fixed geometry, stable lighting). Field robustness under varying illumination, camera motion, and platform re-encoding is an open empirical question.

Unimplemented components. Acoustic encoding, dual-channel binding, and verifier-side bidirectional time bounding are not yet built.

5.4 Open Evaluation Directions

We anticipate empirical evaluation along several axes, with results to be reported in a follow-up paper:

- Bit error rate of optical encoding under varying lighting conditions, and acoustic encoding when implemented.
- End-to-end verification latency on commodity hardware.
- Robustness against re-encoding by commercial video platforms.
- User study of perceptual impact of optical signal embedding.

6 Conclusion

We have presented PoR-Chain, a cryptographic framework for proving that multimedia content was captured at a specific time and through a specific physical capture process. The framework addresses a fundamental gap in existing provenance schemes by binding content to a physical feedback path that no software-only adversary can reproduce, and by anchoring the resulting hash chain to a public ledger in a way that supports efficient verification and tamper-evident sequence integrity. Bidirectional time bounding and dual-channel encoding are designed to extend the framework to defeat increasingly sophisticated synthesis and forgery attacks; acoustic encoding implementing the dual-channel design is planned for future work.

Patent Disclosure

This work is the subject of the following Korean patent applications filed with the Korean Intellectual Property Office (KIPO):

- KR 10-2026-0083791 (Cumulative Hash Chain extension, filed 2026-05-08).

International filings under the Patent Cooperation Treaty are anticipated within the priority window.

References

- [1] S. Kim. Proof of Reality: Binding Multimedia Content to Physical Capture through Ledger-Anchored Feedback Paths. Preprint, KANT Labs, May 2026. <https://doi.org/10.5281/zenodo.20205320>
- [2] Coalition for Content Provenance and Authenticity. C2PA Technical Specification, v1.4, 2024. <https://c2pa.org/specifications/>
- [3] J. Lukas, J. Fridrich, and M. Goljan. Digital camera identification from sensor pattern noise. *IEEE Transactions on Information Forensics and Security*, 1(2):205–214, 2006.
- [4] L. Lamport. Password authentication with insecure communication. *Communications of the ACM*, 24(11):770–772, 1981.
- [5] National Institute of Standards and Technology. Secure Hash Standard (SHS). FIPS Publication 180-4, 2015.
- [6] A. Yakovenko. Solana: A new architecture for a high performance blockchain. Whitepaper, 2018. <https://solana.com/solana-whitepaper.pdf>